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import nltk
nltk.download('brown')
nltk.download('universal_tagset')

[nltk_data] Downloading package brown to /root/nltk_data...
[nltk_data] Unzipping corpora/brown.zip.
[nltk_data] Downloading package universal_tagset to /root/nltk_data...
[nltk_data] Unzipping taggers/universal_tagset.zip.

True

from collections import defaultdict, Counter
from nltk.corpus import brown

sentences = brown.sents()
print("Number of sentences:", len(sentences))

Number of sentences: 57340

tokens = []

for sent in sentences:
    for word in sent:
        tokens.append(word.lower())

print("Total number of tokens:", len(tokens))

Total number of tokens: 1161192

def build_ngram(tokens, n):
    model = defaultdict(Counter)

    for i in range(len(tokens) - n + 1):
        context = tuple(tokens[i:i + n - 1])
        next_word = tokens[i + n - 1]
        model[context][next_word] += 1

    return model

bigram_model = build_ngram(tokens, 2)
trigram_model = build_ngram(tokens, 3)

print("Bigram & Trigram Models are ready")

Bigram & Trigram Models are ready

def predict_next(context, model):
    context = tuple(word.lower() for word in context.split())

    if context not in model:
        return []

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    predictions = model[context].most_common(5)
    return [word for word, count in predictions]

print("Bigram Prediction for 'government':")
print(predict_next("government", bigram_model))

Bigram Prediction for 'government':
['of', '.', ',', 'and', 'was']

print("Trigram Prediction for 'the government decided':")
print(predict_next("the government decided", trigram_model))

Trigram Prediction for 'the government decided':
[]
```